

STRENGTH OF MATERIALS*Time Allowed: 3 Hours**Full Marks: 60***Answer the following questions from Group-A, B & C as directed.****Group A**

1. Choose the correct answer from the alternatives (any ten): 1 x 10
- i. Spring is used to- a) Absorb shock, b) Accelerate speed of a vehicle, c) Both (a) & (b), d) None of the above.
 - ii. The property of a material by virtue of which it can be rolled into plate known as-
a) Ductility b) toughness c) malleability d) none of these.
 - iii. Neutral axis passes through-
a) The C.G. of the section of the beam b) the top layer of the beam
c) The bottom layer of the beam d) none of the above
 - iv. For circular section of diameter 'd', the section modulus is given by _____.
 - v. If diameter of a long column is reduced by 20%, the percentage reduction of Euler buckling load is -
(a) 59, (b) 49, (c) 8, (d) 36.
 - vi. Slope and deflection of a beam can be determined by-
a) Double integration method b) Moment area method
c) Macaulay's method d) any of the above
 - vii. A strut is a member which carries - (a) tensile load, (b) shear load, (c) compressive load, (d) both tensile and compressive load.
 - viii. Poisson's ratio for any material cannot be more than-
a) 0.3, b) 0.5, c) 0.8, d) 1
 - ix. Deformation of a body under tensile load P of cross sectional area A of Length L having Young's modulus E is given by-
a) $\frac{PE}{AL}$ b) $\frac{AE}{PL}$
c) $\frac{PL}{AE}$ d) none of these
 - x. Hooke's law holds good upto the - (a) elastic limit, (b) plastic limit, (c) yield point, (d) proportional limit.
 - xi. Unit of modulus of rigidity - a) N/m^2 b) N/mm^2 c) both a and b d) none of these
 - xii. Bending moment diagram of uniformly distributed force will be-
a) Linear b) Inclined
c) Non-linear d) Horizontal
 - xiii. Percentage of elongation is the measure of-
a) Ductility, b) Malleability, c) Plasticity, d) Elasticity
 - xiv. If diameter of a shaft is doubled the power transmitted capacity will be - (a) same, (b) either twice or half, (c) four times, (d) eight times.

- xv. Shafts are designed on the basis of-
(a) strength (b) rigidity (c) either of the above (d) either (a) or (b)

2. Fill in the blanks (any ten):

1x10

- i. The maximum bending moment (M) caused by the total uniformity distributed load (W) over a simply supported span (L) is given by _____.
- ii. The ratio of crippling load to working load is called _____.
- iii. The angle of twist is _____ proportional to the twisting moment
- iv. Reaction at free end of a cantilever beam is _____.
- v. A beam can be made of uniform strength by keeping the _____ uniform, and varying the _____.
- vi. Unit of radius of gyration is _____.
- vii. The ratio of lateral strain to linear strain is known as _____.
- viii. When a SF at a point is zero, then BM at that point is _____.
- ix. Permanent set occurs when stress is beyond _____ limit.
- x. For a beam with rectangular section, the ratio of maximum shear stress to average shear stress is _____.
- xi. Unit of coefficient of thermal expansion is _____.
- xii. _____ is a graphical method to determine various stresses on an oblique plane of a strained material.
- xiii. The combined effect of external forces acting on a body is called a _____.
- xiv. The strain at right angles to the direction of applied load is known as _____.
- xv. Percentage elongation is a measure of _____.

3. Answer the following questions (any ten):

1x10

- i. Define brittle material.
- ii. What is the helix angle of helical spring?
- iii. Show uniformly varying load on a beam.
- iv. Write the mathematical expression of Temperature stress.
- v. Write down the relation between bulk modulus & compressibility.
- vi. What do you mean by polar modulus?
- vii. What is the expression of Torque transmitted by a solid shaft of diameter d?
- viii. Write the mathematical expression of Polar Modulus of a shaft.
- ix. What is torsional rigidity?
- x. What is the maximum value of S.F, in case of simply supported beam of span l loaded with a point load W at its mid span?
- xi. Describe bending stress.
- xii. What is meant by section modulus?
- xiii. What is law of transmissibility?
- xiv. What is the factor of safety?
- xv. What is SI unit of UDL?

Group B

4. Answer the following questions (any six):

2x6

- i. What is value of bulk modulus of a material having $E=250\text{GPa}$ and $G=80\text{GPa}$?
- ii. Draw a stress strain diagram for ductile materials and indicate all salient features on it.
- iii. Write the equation of torsion and give the names of each terms.
- iv. Define stiffness of a spring.
- v. Write down the difference between deflection & bending moment.
- vi. What is Pure Bending?
- vii. Define SFD and BMD.
- viii. State the perpendicular axis theorem.
- ix. What do you mean by slenderness ratio?
- x. What are the application areas of spring?

Group C

5. Answer the following question (any one): 1x6
- i. What is moment of inertia. What is the difference between mass moment of inertia & area moment of inertia? 2+4
 - ii. Discuss the stress strain behavior of ductile material. Establish the relation $E = 2G(1 + \mu)$ where E is Young's modulus, G is modulus of rigidity and μ is Poisson's ratio. 2+4
 - iii. Find the MI of a Triangular area with respect to the axis passing through the base of the triangle, where, base=b and height =h. 6
6. Answer the following question (any one): 1x6
- i. What are the assumptions made in the Euler's formula? Explain the term effective length of a column. 4+2
 - ii. A cantilever 2m long carries point loads of 4kN, 5kN, 3kN at distances 1m, 1.5m, and 2m respectively from the fixed end sides. Draw the shear force and bending moment diagram for the beam. 6
 - iii. A T beam 200mm*150mm*10mm is used over a span of 3 meter It carries a uniformly distributed load of 12000 N per meter run. Find the position of the neutral axis and the maximum stress induced in the beam due to bending. Also state the nature of maximum bending stress. 6
7. Answer the following question (any one): 1x6
- i. A solid circular shaft of 100 mm diameter is transmitting 120 kW at 150 r.p.m. Find the intensity of shear stress in the shaft. <https://www.wbscteonline.com> 6
 - ii. What do you mean by equivalent stiffness of helical spring? Draw sketches showing series connection and parallel connection of helical springs. 2+4
 - iii. A beam of 3m span is loaded as shown in the fig 6. Calculate the deflection at P and Q and slope at A. Take the value of $EI = 840 \text{ KN-m}^2$ 6

